Norms and Geometry: Why L^2 Isn't Always the Answer

🤔 **Student Question**

From the class: "A norm just measures the 'size' of a vector. The Euclidean norm (L^2) is the natural one from geometry—it's just the straight-line distance. Can't we use it everywhere?"

This seems obvious—L^2 is how we measure distance in the physical world!

The Natural Approach: Always Use Euclidean (L^2) Norm

The Euclidean norm feels natural because it matches our geometric intuition:

Definition: $\|x\|_2 = \sqrt{\sum_{i=1}^{n} x_i^2}$

Why it seems universal:
- Measures straight-line distance (Pythagorean theorem)
- Rotation-invariant (all directions treated equally)
- Has nice inner product structure: $\|x\|_2^2 = \text{inner}(x, x)$
- Used in physics, engineering, everyday geometry

Applications in ML:
- Ridge regression: minimize $\|y - X\theta\|_2^2 + \lambda \|\theta\|_2^2$
- Neural networks: minimize $\|y - f(x; \theta)\|_2^2$
- K-means: minimize $\sum_i \|x_i - \mu_{c_i}\|_2^2$

So why consider other norms?

Where L^2 Breaks: Sparse vs Dense Solutions

Let's see what happens with different norms in a concrete optimization problem.

Problem: Minimize $f(x) + \text{regularization penalty}$

Setup:
- Loss function: $f(x) = (x_1 - 100)^2 + (x_2 - 1)^2 + (x_3 - 1)^2 + \ldots + (x_{100} - 1)^2$
- Constraint: $\|x\| \leq 100$ (regularization)

Intuition: First coordinate "wants" to be 100, others want to be 1.

With L^2 norm constraint: $\|x\|_2 \leq 100$

Optimal solution: Distribute budget across all coordinates
- $x^* \approx (99.5, 0.995, 0.995, \ldots, 0.995)$ (all coordinates non-zero)
- Result: **Dense solution** (100 non-zero entries)

With L^1 norm constraint: $\|x\|_1 \leq 100$

Optimal solution: Concentrate budget on important coordinates
- $x^* = (100, 0, 0, \ldots, 0)$ (only first coordinate non-zero)
- Result: **Sparse solution** (1 non-zero entry!)

The difference: Same constraint budget (100), completely different solutions!

⚠️ **Crisis**

The Norm Choice Crisis: Different Solutions from Different Norms

Problem: Optimization with norm constraint

What happens:
- L^2 norm: Encourages dense solutions (all coordinates contribute)
- L^1 norm: Encourages sparse solutions (few coordinates non-zero)
- $L\infty$ norm: Bounds maximum coordinate (different structure)

Concrete example: Problem: Fit model with 100 features, constraint budget = 100

L^2 regularization (Ridge):
- Solution: All 100 features have small weights
- Example: $\theta = (0.5, 0.3, 0.2, 0.1, 0.05, ...)$ (all non-zero)
- Interpretation: All features matter a little

L^1 regularization (Lasso):
- Solution: Only 10 features have non-zero weights
- Example: $\theta = (2, 0, 0, 3, 0, 0, 1, 0, ...)$ (90% zeros!)
- Interpretation: Feature selection—only important features

The gap: 100 features vs 10 features (10× sparsity!)

Impact: Norm choice is an inductive bias, not just a technicality

Understanding Why: Geometry of Norm Balls

The reason lies in the geometry of different norm balls.

Unit balls: $\{x : \|x\| \leq 1\}$

📊 Comparison

Geometry of ℓ_p norms in $\mathbb{R}^2$:

L¹ norm: $\|x\|_1 = |x_1| + |x_2|$
- Unit ball: Diamond shape
- Corners at axes: $(\pm 1, 0)$ and $(0, \pm 1)$
- Property: Optimal points tend to lie on axes (sparsity!)

L² norm: $\|x\|_2 = \sqrt{x_1^2 + x_2^2}$
- Unit ball: Circle
- No special directions—all treated equally
- Property: Smooth boundary, no sparsity preference

L∞ norm: $\|x\|_\infty = \max(|x_1|, |x_2|)$
- Unit ball: Square
- Corners at diagonals: $(\pm 1, \pm 1)$
- Property: Bounds maximum coordinate

Why sparsity happens with L¹: When a level set of the objective function (ellipse for quadratic) touches the L¹ ball, it typically hits a corner—which lies on an axis (sparse!)

For L² ball (circle), contact point is smooth—no preference for axes

The Key Insight: Norm Choice Encodes Prior Knowledge

🎯 Key Takeaway

Understanding Norm Choice as Inductive Bias

Different norms encode different assumptions about solutions:

- **L¹:** "Solution should be sparse" (few non-zero features)
- **L²:** "Solution should have small, distributed weights" (all features contribute)
- **L∞:** "No single feature should dominate" (bounded max)

Norm choice determines solution structure, not just size!

Definition: Norm

A **norm** on vector space V is a function $\|\cdot\| : V \to \mathbb{R}_+$ satisfying:
1. **Positivity:** $\|x\| \geq 0$ with equality iff $x = 0$
2. **Homogeneity:** $\|\alpha x\| = |\alpha| \, \|x\|$ for scalars α
3. **Triangle inequality:** $\|x + y\| \leq \|x\| + \|y\|$

Intuition: A norm assigns "size" to vectors, generalizing length

Definition: ell_p Norms

For $x \in \mathbb{R}^n$ and $p \geq 1$:

$$\|x\|_p = \left(\sum_{i=1}^{n} |x_i|^p\right)^{\frac{1}{p}}$$

Special cases:

- $\|x\|_1 = \sum_{i=1}^{n} |x_i|$ (sum of absolute values)
- $\|x\|_2 = \sqrt{\sum_{i=1}^{n} x_i^2}$ (Euclidean distance)
- $\|x\|_\infty = \max_i |x_i|$ (maximum absolute value)

Applications in Machine Learning

📊 **Comparison**

Norm choice in regularization:

Method	Penalty	Solution Type	Use Case
Ridge	$\lambda \ \|\theta\|_2^2$	Dense	Numerical stability
Lasso	$\lambda \ \|\theta\|_1$	Sparse	Feature selection
Elastic Net	$\lambda_1 \ \|\theta\|_1 + \lambda_2 \ \|\theta\|_2^2$	Sparse + stable	High-dimensional data
Max-norm	$\lambda \ \|\theta\|_\infty$	Bounded max	Robust to outliers

Example (gene expression data):

- 20,000 genes (features), 100 patients (samples)
- L^2 (Ridge): Uses all 20,000 genes with small weights (hard to interpret)
- L^1 (Lasso): Selects 50 genes with non-zero weights (interpretable!)
- **Sparsity: 20,000 $\longrightarrow$ 50 features (400× reduction!)**

Example: Ridge vs Lasso Regression

Ridge regression:

$$\min_{\theta} \|y - X\theta\|_2^2 + \lambda \|\theta\|_2^2$$

Solution: $\left(X^T X + \lambda I\right)^{-1} X^T y$ (closed form!)

- All coefficients shrunk toward zero
- No coefficients exactly zero
- Good for prediction, hard to interpret

Lasso regression:

$$\min_{\theta} \|y - X\theta\|_2^2 + \lambda \|\theta\|_1$$

Solution: No closed form, use iterative methods

- Many coefficients exactly zero
- Automatic feature selection
- Interpretable models

Real example (Boston housing):

- 13 features, 506 samples
- Ridge: All 13 features non-zero
- Lasso (λ=0.1): Only 8 features non-zero
- **Automatic feature selection via sparsity!**

Why Norm Equivalence Doesn't Save Us

You might think: "In finite dimensions, all norms are equivalent, so they're basically the same!"

Theorem: Norm Equivalence

In $\mathbb{R}^n$, for any norms $\|\cdot\|_a$ and $\|\cdot\|_b$, there exist $c, C > 0$ such that:

$$c \|x\|_a \leq \|x\|_b \leq C \|x\|_a \quad \text{for all } x$$

Why this doesn't mean norms are interchangeable:

1. **Constants can be large**
   - In $\mathbb{R}^{1000}$: $\|x\|_2 \leq \|x\|_1 \leq \sqrt{1000} \|x\|_2$
   - $\sqrt{1000} \approx 31.6$ gap makes bounds very loose!
2. **Optimization favors different solutions**
   - Equivalent topologically (same convergence)
   - Not equivalent for optimization (different optima!)
3. **Inductive bias matters**
   - L^1 sparsity is a feature, not a bug
   - We choose norms to encode prior knowledge

🎯 **Key Takeaway**

Norm equivalence means:
- Convergence in one norm ⟶ convergence in all norms
- All norms induce same topology (open sets)

Norm equivalence does NOT mean:
- Optimization gives same solutions
- Regularization has same effect
- Geometric properties are identical

Bottom line: Norm choice encodes inductive bias about solution structure

Inner Products and Angles

Some norms come from inner products, which add geometric structure:

Definition: Inner Product

An **inner product** $\mathrm{inner}(\cdot, \cdot)$ satisfies:
1. Linearity: $\mathrm{inner}(\alpha x + \beta y, z) = \alpha\ \mathrm{inner}(x, z) + \beta\ \mathrm{inner}(y, z)$
2. Symmetry: $\mathrm{inner}(x, y) = \mathrm{inner}(y, x)$
3. Positive-definiteness: $\mathrm{inner}(x, x) \geq 0$ with equality iff $x = 0$

Induced norm: $\|x\| = \sqrt{\mathrm{inner}(x, x)}$

Key property: L^2 norm comes from standard inner product $\mathrm{inner}(x, y) = \sum_i x_i y_i$

But: L^1 and $L\infty$ norms do NOT come from inner products!
- Can't define angles with L^1 norm
- Can't project orthogonally with $L\infty$ norm

Impact on ML:
- PCA requires inner product (uses L^2)
- Kernel methods require inner product structure
- Some algorithms need L^2 for geometric operations

Cauchy-Schwarz: The Fundamental Bound

For norms induced by inner products, we have:

Theorem: Cauchy-Schwarz Inequality

For vectors x, y in an inner product space:

$$|\mathrm{inner}(x, y)| \leq \|x\|\ \|y\|$$

with equality iff x and y are linearly dependent

Proof sketch: Consider $f(t) = \|x - ty\|^2 \geq 0$ for all t. Expand: $f(t) = \|x\|^2 - 2t\ \mathrm{inner}(x, y) + t^2\ \|y\|^2$ This quadratic is always non-negative, so discriminant ≤ 0:

$$4\mathrm{inner}(x, y)^2 - 4\|x\|^2\ \|y\|^2 \leq 0$$

Applications:
- Proves triangle inequality
- Bounds correlations: $|\text{Corr}(X, Y)| \leq 1$
- Controls Rademacher complexity (Week 3)

🔮 **Looking Ahead**

We've seen that norm choice determines solution structure in finite dimensions.

But what about infinite-dimensional spaces? (Function spaces)

In finite dimensions: All norms equivalent (same topology) In infinite dimensions: **Norms NOT equivalent!** Different convergence notions

Brief preview: Functional analysis (next section) handles infinite dimensions, where norm choice becomes critical (L^2 vs L^1 vs $L\infty$ for functions give different spaces!)

Summary: Choosing the Right Norm

🎯 **Key Takeaway**

When to use each norm:

L^1 (Lasso):
- Want sparse solutions (feature selection)
- High-dimensional data (p >> n)
- Interpretability important
- Example: Gene selection, image compression

L^2 (Ridge):
- Want all features to contribute
- Numerical stability important
- Natural for Euclidean geometry
- Example: Ridge regression, neural networks

$L\infty$ (Max-norm):
- Want bounded maximum
- Robustness to outliers
- Minimax optimization
- Example: Robust statistics, game theory

Elastic Net (L^1 + L^2):
- Want sparsity AND stability
- Correlated features
- Best of both worlds

The principle: Norm is not just technicality—it's a design choice encoding prior knowledge about solutions!